Ashutosh Kumar

Roll No.: 220108014
B.Tech - Electronics and Electrical Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	6.1	2022–2026
Senior Secondary	CBSE Board	84.0%	2022
Secondary	CBSE Board	93.4%	2020

Projects

- **Ball Balancing System using PID** *Jun. 2025*
  Group Project Github
  - Designed and implemented an advanced ball balancing system controlled by a **PID algorithm** on **Arduino microcontroller**.
  - Utilized dual **ultrasonic sensors** with **Kalman filtering** for highly accurate distance measurement and effective noise reduction.
  - Integrated **precision servo motor actuation** to dynamically stabilise ball position on the custom balancing platform.
  - Conducted a comprehensive performance analysis by tuning **PID gains** to minimize **steady-state error** and **overshoot**.
- **Smart Dual Axis Solar Tracking System** *Jul. 2025*
  Group Project Github
  - Developed a dual axis solar tracking system using **Arduino Uno R3** to maximize solar panel energy efficiency.
  - Implemented robust**LDR-based sensing mechanism** for precise sunlight detection and automated solar panel alignment.
  - Controlled **precision servo motors** for **real time elevation** and **azimuth** adjustments ensuring efficient **sun tracking** performance.
  - Designed and optimized intelligent system with **tolerance-based dynamic error correction**, enhancing overall stability and tracking accuracy.
- **Ultrasonic Mini Radar System** *Apr. 2025*
  Guide: Prof. Ashwini Sawant | EEE, IIT Guwahati Github
  - Developed an **Ultrasonic Mini Radar System** using Arduino with servo-based angular scanning and **real-time object tracking**.
  - Implemented **ultrasonic sensing, LCD, serial monitor** and buzzer for accurate distance detection and obstacle warning.
  - Integrated RGB LED indicators with motion analysis to classify **approaching**, **moving away** or **stationary** object.

Technical Skills

- **Programming Languages:** C/C++
- **Operating System:** Windows, MacOS
- **Simulation:** Tinkercad–Circuit design and simulation
- **Miscellaneous:** Arduino IDE

Achievements

- **Government of India:** Official invitee to Rashtrapati Bhavan for an interactive session with the President of India. *2026*
- **Juspay Hiring Challenge:** Secured a place among top 500+ out of 2,00,000+ participants nationwide. *2025*
- **Meesho Dice Challenge 2.0 – Tech Track:** Achieved a rank of 215 among 2600+ participants nationwide. *2025*
- **Reliance Foundation Scholar:** Selected among 40,000+ applicants nationwide. *2022*
- **NTSE Qualified:** National Talent Search Examination (NCERT, New Delhi). *2020*
- **District-Level GK Competition:** Secured Rank 1 among 2000+ participants. *2018*

Key courses taken

- **Electrical**: Principles of Electrical Engineering, Electrical Machines, Power electronics, Electrical Power systems.
- **Electronics**: Control Systems, Digital Circuits, Embedded Systems, Control and Robotics Laboratory.
- **Programming**: Introduction to Computing, Data Structures and Algorithms.

Positions of Responsibility

- **Core Team Member:** LitSoc – Literary Club, IIT Guwahati *2024*
  Organized literary events such as *Akhar* – the annual literary fest of IITG, and *Ramcharit* – a discussion series on the Ramayana.
- **City Representative:** Technothlon, Techniche, IIT Guwahati *2023*

Extracurricular Activities

- **Inter IIT Cultural Meet:** Represented IIT Guwahati in Hindi Poetry at the 7th edition hosted by IIT Patna. *2024*
- **Alcheringa :** Associate Executive, CRM team at IIT Guwahati's annual cultural fest. *2024*
- **Manthan :** Represented Brahmaputra Hostel at the Inter-Hostel Cultural Competition, IIT Guwahati. *2023*